Review plan · for the teacher

Smart school projects: unit check

IoT · Level 4 · Unit check

Time	Materials
45 min	Board, chalk, paper. No devices needed.

Learning goals

- Start from a problem
- Pilot, place and test
- Keep it running and secure
- Data ethics: fair, safe and honest

Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Write the hardest question on the board for tomorrow

Discussion

- Which idea will you use first?
- Which question surprised you, and why?
- What will your pupils build or measure first?

Exit ticket

- One idea I will use, and when I will use it

Try it at school tomorrow

Pick the idea from this unit your pupils need most, teach it this week, and tell a colleague how it went.

Quiz sheet · for each team

Smart school projects: unit check

IoT · Level 4 · Unit check

Team _____ Date _____

1 **Which is the best success measure for the water-tank project?**

- ◯ No dry days by the end of term (five last term)
- ◯ Fewer days without water than before
- ◯ Three sensors installed and online by the end of March

2 **A CO_2 sensor beside an open window usually reads lower than the air most pupils breathe.**

- ◯ True
- ◯ False

3 **Match each practice to the problem it prevents.**

A measurable question	• • • •	The same mistake built ten times
A pilot node	• • • •	A project nobody can judge
A heartbeat	• • • •	People surprised by what's collected
Asking families first	• • • •	A dead sensor that still looks fine

4 **For the tank, a miss means a day without water, and a false alarm means a wasted climb up the tower. Where should the low-level alert sit?**

- ◯ It doesn't matter much: both mistakes cost about the same
- ◯ Very low: alert only when nearly empty, so there are no false alarms
- ◯ Early: accept some false alarms so real low levels aren't missed

5 **Two AA batteries are rated 2,500 mAh, and the node averages 1.75 mA. Plan with 70% of the label.**

Usable capacity: _____ mAh. Battery life: about _____ hours, roughly six weeks.

Word bank: 2,500 · 1,750 · 1,429 · 1,000

6 **Which are good security habits for school sensors?**

- ☐ Leave the factory password, so the supplier can help
- ☐ Give each device its own new password
- ☐ Keep the sensors on their own network
- ☐ Turn on every feature, in case it's useful
- ☐ Install updates when they come

7 **Put a school sensor project in order.**

- [ ] Pilot one node with a written test
- [ ] Hand over: a guide, the passwords, trained people
- [ ] Build, place and set the alerts
- [ ] Find a problem people feel
- [ ] Write a measurable question and a success measure

8 **Data from a classroom CO_2 sensor is about air, so it can't tell you anything about people.**

◯ True ◯ False

9 **Room 6 went above 1,500 ppm on 7 of the 8 afternoons measured, with one sensor. Which is the most honest headline?**

◯ Room 6's teacher keeps the windows shut in the afternoons

◯ Room 6 was stuffy on most afternoons we measured

◯ Room 6 has the worst air of any classroom in the school

10 **Which should the families' notice for a CO_2 project say?**

- [ ] Every raw reading from every room
- [ ] Which teachers' rooms were worst
- [ ] How long it is kept
- [ ] Whom to ask with questions
- [ ] What is measured, and why
- [ ] Who can see the data

Answers · for the teacher

Smart school projects: unit check

IoT · Level 4 · Unit check

1 **No dry days by the end of term (five last term)**

It names the result the school wants, with a number, a date and a baseline to compare with.

2 **True**

Outdoor air, about 420 ppm, reaches it first, while the middle of a full room can be far higher.

3 **A measurable question — A project nobody can judge**
A pilot node — The same mistake built ten times
A heartbeat — A dead sensor that still looks fine
Asking families first — People surprised by what's collected

One practice from each lesson: plan with a question you can check, test one before many, notice silent failures, and ask before you collect.

4 **Early: accept some false alarms so real low levels aren't missed**

A miss costs far more than a false alarm here, so the threshold should lean towards alerting early, leaving time to refill.

5 **Usable capacity: 1,750 mAh. Battery life: about 1,000 hours, roughly six weeks.**

70% of 2,500 is 1,750 mAh, and 1,750 ÷ 1.75 = 1,000 hours, about 42 days.

6 **Give each device its own new password; Install updates when they come; Keep the sensors on their own network**

Mirai got in with factory passwords, so each device needs its own; updates close known weaknesses; and a separate network limits the harm.

7 **Top to bottom: 3, 5, 4, 1, 2**

This is the course in one line: problem, question, pilot, build, hand over.

8 **False**

CO_2 rises when people are in the room and falls when they leave, so the data shows when rooms are full or empty, and with a timetable, which teacher was where.

9 **Room 6 was stuffy on most afternoons we measured**

It says what was measured and when, and no more; the report can then add the sensor and the dates.

10 **What is measured, and why; Who can see the data; How long it is kept; Whom to ask with questions**

A good notice answers what, why, who sees it, how long, and whom to ask, in plain words.